ZOO506

IMPORTANT SHORT QUESTION

Q. Damage cause by termites.

They enter through cut ends of setts & feed on the soft tissue. The tunnel is filled with the soil.

The termites attack setts, shoots, canes and also stubbles Entire shoot dries up and can be pulled out easily

Q. Intelleigent cultivation.

- Intelligent cultivation and use of fertilizers is well as pruning to remove injured or diseased parts will aid in insect control

Q.what are repellents? Explain some repellents.

1-Repellents are those materials used to keep insects away from crops, animals and man

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2. Various coal-tar and pine products are rather general insect repellents.

3. Naphthalene is a clothes moth repellent and oil of citronella is a common ingredient of preparations used to keep mosquitoes away from people

Q .write commercial products produce by insects..

A third group of insects is responsible for a number of commercial products

1-Silk

2-Honey

3-Beeswax

4-Shellac

Q. Write the uses of pesticides.

Pesticides are used to control various pests and disease carriers, such as mosquitoes, ticks, rats and mice. Pesticides are used in agriculture to control weeds, insect infestation and diseases. There are many different types of pesticides; each is meant to be effective against specific pests

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Q. Mode of damage of fruit borer?

Bolls showing regular, circular bore holes Larvae seen feeding on the boll by thrusting their heads alone inside and leaving the rest of the body outside

Presence of granular faecal pellets outside the bore hole.

A single larva can damage 30-40 bolls

Q. Write the name of thorax seqments.

The thorax has its three segments usually quite clearly marked 1-Prothorax

2-Mesothorax

3-Metathorax

Q. Vector of malaria?

In the case of malaria, the vector is the anopheline mosquito and the disease-causing organism is the

malaria parasite.

Q. What is shell gland?

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A gland-producing material, which forms the egg shell and is known as the shell gland, also opens into .this portion

Q. Example of hemiptera

Examples: Assassin bugs, Kissing bugs, Leaffooted bug, Lygus bug, Stink bug, Minute piratenbug,Big-eyed bug, Damsel bug

Q. Phthiraptera

Formally Anoplura and Mallophaga

Includes: Lice, Chewing Louse, Sucking Louse

Mouthparts: Chewing and Piercing-Sucking

Metamorphosis: Complete Metamorphosis

Order : Psocodea

Note: order or example examiner poch rha hy.

Q. Transmission

Transmission is the passing of a pathogen causing communicable disease from an infected host individual or group to a particular individual or group .Regardless of whether the other individual was previously infected

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Q. Coxa

The coxa is the segment that articulates with the body, frequently partly lying in a more or less cupshaped hollow of the latter

It may be short or long and is generally freely movable on the body and powerful

Q. Cause of allergy

Allergy (hay fever, asthma, etc.) are frequently caused by exposure of susceptible persons to dried

insect scales and hairs

Q. Classes of Arthropods

1. Class #1- Hexapoda: Insects, six, jointed appendages

2. Class #2- Chilopoda- Centipedes, one pair of legs per body segment, multiple body segments

3. Class #3- Diplapoda- Millipedes, two pairs of legs per body segment, multiple body segments

4. Class #4- Crustacea- Crayfish, five pairs of jointed appendages

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Q. Epidemiology

The study of disease in populations of an organism.

Q. Where live grasshopper of rice?

Various species of grasshopper are widely distributed in Nepal They are polyphagous and feed on leaves of rice, maize, millets, sugarcane, grasses etc

Q Fuzz purposes

Fuzz: It is used in production of mattresses, surgical cotton, photographic film and paper.

Q. Scientific name of early shoot borer

For Early shoot borer :

Trichogramma chilonis

Q.What is Biological Control?

Manipulation of biological organism to control pests by:

1- Release of predators/parasites/disease of an insect or weed Can be time consuming, expensive and difficult

Q.What is Host Plant Resistance?

Manipulating the crop to withstand or tolerate pests by

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1- Natural breeding method

2- Genetically modified plants

Examples: Glandular-haired Alfalfa, Bt Corn,

Q.where Rice Grass hopper live 2

Various species of grasshopper are widely distributed in Nepal

They are polyphagous and feed on leaves of rice, maize, millets, sugarcane, grasses etc

Q.Which machinery is used to control pest 2

Uses machinery and/or other tools to control pests

1- Tillage: Tillage is the agricultural preparation of soil by mechanical agitation of various types, such as digging, stirring, and overturning

Q.Symptoms of damage of Fruit borer 3

Bolls showing regular, circular bore holes

Larvae seen feeding on the boll by thrusting their heads alone inside and leaving the rest of the body outside

Presence of granular faecal pellets outside the bore hole.

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A single larva can damage 30-40 bolls.

Q.Why Agronomic practices that are designed to: 3

1-Optimize growing conditions for the crop

2-Anything that increases a crop's competitive edge will result in increased tolerance to pests often resulting in reduced pesticide use

Q- Create unfavorable conditions for the pest ?

1- Describe the places where heat treatment is used to kill the insects 3

2- Most insects cease reproduction and soon die at temperatures rom 100 to 1l0°F

3- The temperature of their surroundings should be raised to 120° or higher to kill them with heat artificially within a few hours

4- The length of period for effective treatment is more dependent upon the

a. time to heat the grain

b. clothing

c. other material in which the insects are located

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5- Than upon the heat resistance of the pests themselves

6- Because no insect can survive more than an hour if actually exposed to a temperature of 120°F 7- Bales and bags are difficult to heat through

Q. Define Ovoviviparous.

Those having eggs that remain in the body of the parent for a time after hatching are termed ovoviviparous.

Q. What is meant by contact insecticides?

It was formerly supposed that certain materials called contact insecticides, which kill insects by contact with their bodies. It is caused death by entering the spiracles and closing them up, thus producing suffocation.

Q: What is seminal receptcal?

A side pouch called seminal receptacle is connected with the vagina is for the storage of the sperms which fertilize the eggs.

Q: Write down the scientific name of early shoot borer. Trichogramma chilonis is the scientific name of early shoot borer.

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Q: Define Arthopods.

The largest group of all, however, is the Arthropoda. Its members found in the seas, in fresh water, on land or even flying freely. A group with remarkable differences of structure and it is so abundant that all the other animals taken together are less than one-sixth as many as the arthropods.

Q: What is Biological Control?

Manipulation of biological organism to control pests by:

1- Release of predators/parasites/disease of an insect or weed

2- Can be time consuming, expensive and difficult

Q: Write down the Order of termites.

Blattodea is the Order of termites.

Q: What is the appearance of adult rice-ear?

The adult of Leptocorisa oratorius is green, light brown or mixed yellow in color.

Q: Write down the economically importance of insects.

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1. Of the many thousands of insect species known to man some affect him and his property only slightly; some are beneficial; others are injurious in various degrees.

Q: Explain Positions Head of insects and its types.

The heads of different insects very much in form in the location of the mouth. In some cases this is on the underside; in others it is practically on the front.

Hypo-gnathous head perpendicular to longitudinal body axis for example Locust.

Pro- gnathous head extended to longitudinal body axis for example Weevils.

Opitho- gnathous head bent underneath thorax during rest for example Aphids.

Q. List the insect born diseases?

- Malaria
- Bubonic plague
- Typhoid
- Typhus Yellow fever

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Q. Clean culture?

The destruction of vegetable trash and weeds is known as cultural clean

Q. Hopper burn cause?

Brown plant hopper is the most destructive pest of rice .. both nymph and adult feed on paddy, sugercan and grasses by sucking cell sap which turn the leaves yellow.

A heavy infestation produce symptoms of “hopper burn” i.e. leaves become dry and brown after insect feeding and patched of burned plants are often lodged.

Q.Damage cause by termits?

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• The tunnel is filled with the soil.

• The termites attack setts, shoots, canes and also stubbles

• Entire shoot dries up and can be pulled out easily

Q. Physical control methods?

Physical factor controlling insects are following

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- Heat
- Cold
- Light
- Moisture
- Electricity

Q. Enlist the plants require pollination?

Smyrna fig, clovers, melons, tomatoes, beans, peas,and most of common fruits require insects for their pollination

Q. Agronomic practice in cultural control?

1. Optimize growing conditions for the crop

2. Anything that increases a crop's competitive edge will result in increased tolerance to pests often resulting in reduced pesticide use

3. Create unfavorable conditions for the pest

Q. What is cotton?

Cotton is a soft, fluffy staple fiber that grows in boll, or protective Cotton (Gossypium spp), the king of fibers, usually commercial referred as white gold and one of the important commercial crops, plays a pivotal role in

human civilization, economic, political and social affairs of world.

Q. Economic threshold?

“Pest Population at which a grower must take action to prevent a pest populations fromreaching the economic injury level”

Q. Define pulvilus, pulvilli,empodium?

At the end of the last a pair of claws is generally found, and between them a sort of pad or cushion, the pulvillus

Sometimes there are three of these, in which case the outer ones are called the pulvilli and the middle one the empodium

Q. Intelligent cultivation?

Intelligent cultivation and use of fertilizers is well as pruning to remove injured or diseased parts will aid in insect control

Q. Arthropoda classes?

5 Classes of Arthropods

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4. Class #4- Crustacea- Crayfish, five pairs of jointed appendages

5. Class #5- Arachnida- Spiders, four pairs of jointed appendages

Q. Insect pest management?

The stud of principles and experimental approaches to the science that underpins the development of working integrated pest management systems.

Q.Appearance of adult rice-ear.2

Rice ear- head bug

The adult of Leptocorisa oratorius is green, light brown or mixed yellow in color The adult are slender and about 20mm long.

Q. Define arthropods? 2

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An “arthropod” is an invertebrate animal that has an exoskeleton, a segmented body, and jointed appendages any of a phylum (Arthropoda) of invertebrate animals (such as insects, arachnids, and crustaceans) that have a segmented body and jointed appendages, a usually chitinous exoskeleton molted at intervals, and a dorsal anterior brain connected to a ventral chain of ganglia

Q.Damage by termites?3

Termite

SYMPTOMS OF DAMAGE

They enter through cut ends of setts & feed on the soft tissue.

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Q. What are attractant? 3

Attractants

1. Sometimes called attrahents are materials used to lure insects to traps and poisoned baits

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2. The use of geraniol in Japanese beetle traps, sugar sirups for ant baits, and fermenting sirup solutions for trapping various moths are good examples.

Q. What is biological control?

Biological control or biocontrol is a method of controlling pests such as insects, mites, weeds and plant diseases using other organisms. It relies on predation, parasitism, herbivory, or other natural mechanisms, but typically also involves an active human management role.

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